

NetPath

High-Level Use Cases Document

Based on the Proof of Concept for the NetPath interactive network education platform

Version: 1.0 | Date: May 2026 | Scope: Main product use cases for learners, educators, team administrators, and supporting external systems.

1. Purpose and Scope

This document identifies the high-level actors and main use cases for NetPath, a browser-based, sandbox-driven platform that teaches networking fundamentals through guided tutorials, drag-and-drop topology building, validation feedback, request simulation, progress tracking, and classroom or corporate cohort features.

The use cases focus on conceptual platform behavior rather than implementation details. Low-level packet timing, packet loss, and performance modeling are out of scope for the initial product concept.

Product Area	Relevant Proof-of-Concept Capability
Guided Tutorial Stages	Learners progress through short stages that explain, demonstrate, practice, and validate concepts.
Interactive Sandbox	Learners drag devices to a canvas, draw wired or wireless connections, and configure device settings such as IP addresses, gateways, or DHCP.
Validation Engine	The system checks topology reachability and stage success criteria in real time using graph traversal.
Request Simulation	Learners send a simulated request and watch the request/response path through the network.
Progression and Gamification	Stage unlocks, badges, topology portfolio, timed challenges, leaderboards, and peer sharing support engagement.
Institutional and Team Features	Teams and academic customers can use admin dashboards, cohort progress tracking, assignment modes, LMS integrations, and custom challenges.

2. Actors

2.1 Primary Actors

Actor	Description	Primary Goals
Learner	A student, self-taught developer, career-switcher, or employee using NetPath to learn networking.	Complete tutorials, build networks, validate designs, simulate requests, save work, and track progress.
Instructor / Educator	A teacher, professor, bootcamp instructor, or training facilitator using NetPath with a class or cohort.	Assign lessons or challenges, monitor learner progress, review submissions, and integrate with classroom workflows.
Team Administrator	A corporate training, HR/L&D, bootcamp, or institutional administrator managing seats and cohorts.	Invite users, manage seats and billing, view team analytics, and configure team-level learning experiences.
Curriculum / Platform Administrator	An internal NetPath administrator or content owner responsible for product content and platform configuration.	Create and publish modules, manage device unlocks, define validation rules, review analytics, and maintain platform quality.

2.2 Secondary Actors and Supporting Systems

Actor / System	Type	Role in Use Cases
Authentication / SSO Provider	External system	Authenticates users via social login, SSO, or enterprise SAML.
Payment Processor	External system	Processes Pro, Teams, and Academic subscriptions.
Learning Management System (LMS)	External system	Receives assignments, progress, and roster data through an LTI-style integration.
Email / Notification Service	External system	Sends onboarding, challenge, retention, and progress notifications.
Database / Progress Store	Internal supporting system	Persists learner accounts, progress, portfolios, badges, cohorts, and subscriptions.
Simulation and Validation Engine	Internal supporting system	Checks topology correctness and animates request paths for learner feedback.

3. Use Cases by Actor

Actor	ID	Use Case	High-Level Description
Learner	UC-L01	Start Guided Tutorial	Begin a lesson stage, view explanatory content, and practice the concept in a prepared environment.
Learner	UC-L02	Register / Sign In	Create an account or authenticate to save progress and access paid or assigned content.
Learner	UC-L03	Build Sandbox Topology	Drag devices onto the canvas and connect them using Ethernet or wireless associations.
Learner	UC-L04	Configure Devices	Open a configuration panel and set IP addresses, gateways, DHCP, or lesson-specific settings.
Learner	UC-L05	Validate Topology	Receive real-time green, amber, or red feedback against lesson success criteria.
Learner	UC-L06	Simulate Request Path	Send a simulated HTTP request and watch the request and response move through the topology.
Learner	UC-L07	Save / Revisit Portfolio	Save completed sandbox designs and revisit or reuse them later.
Learner	UC-L08	Attempt Timed Challenge	Complete optional timed challenges for advanced practice or certification preparation.
Learner	UC-L09	Share Network Design	Generate a shareable visual card or classroom submission from a completed network.
Learner	UC-L10	Upgrade to Pro	Access WAN content, additional stages, unlimited saves, certificates, and offline mode.
Instructor / Educator	UC-E01	Create / Manage Cohort	Create a class, invite learners, and organize them into groups.
Instructor / Educator	UC-E02	Assign Lessons and Challenges	Assign specific stages or sandbox challenges to learners.
Instructor / Educator	UC-E03	Review Progress Analytics	View completion, stage performance, badges, and learner engagement.
Instructor / Educator	UC-E04	Use LMS Integration	Sync rosters, assignments, and progress with Canvas, Moodle, or Google Classroom.
Instructor / Educator	UC-E05	Review Shared Designs	Open learner-submitted topology designs for assessment or feedback.
Team Administrator	UC-T01	Invite Team Members	Add learners to a team account and manage access.
Team Administrator	UC-T02	Manage Seats and Billing	Purchase, assign, remove, or renew seats for Teams or Academic tiers.
Team Administrator	UC-T03	Create Custom Challenge	Create or configure challenges aligned with internal onboarding or training needs.
Team Administrator	UC-T04	Track Team Progress	Review team-level progress, engagement, and completion metrics.
Team Administrator	UC-T05	Request Demo / Pilot	Submit interest for a B2B pilot or institutional evaluation.

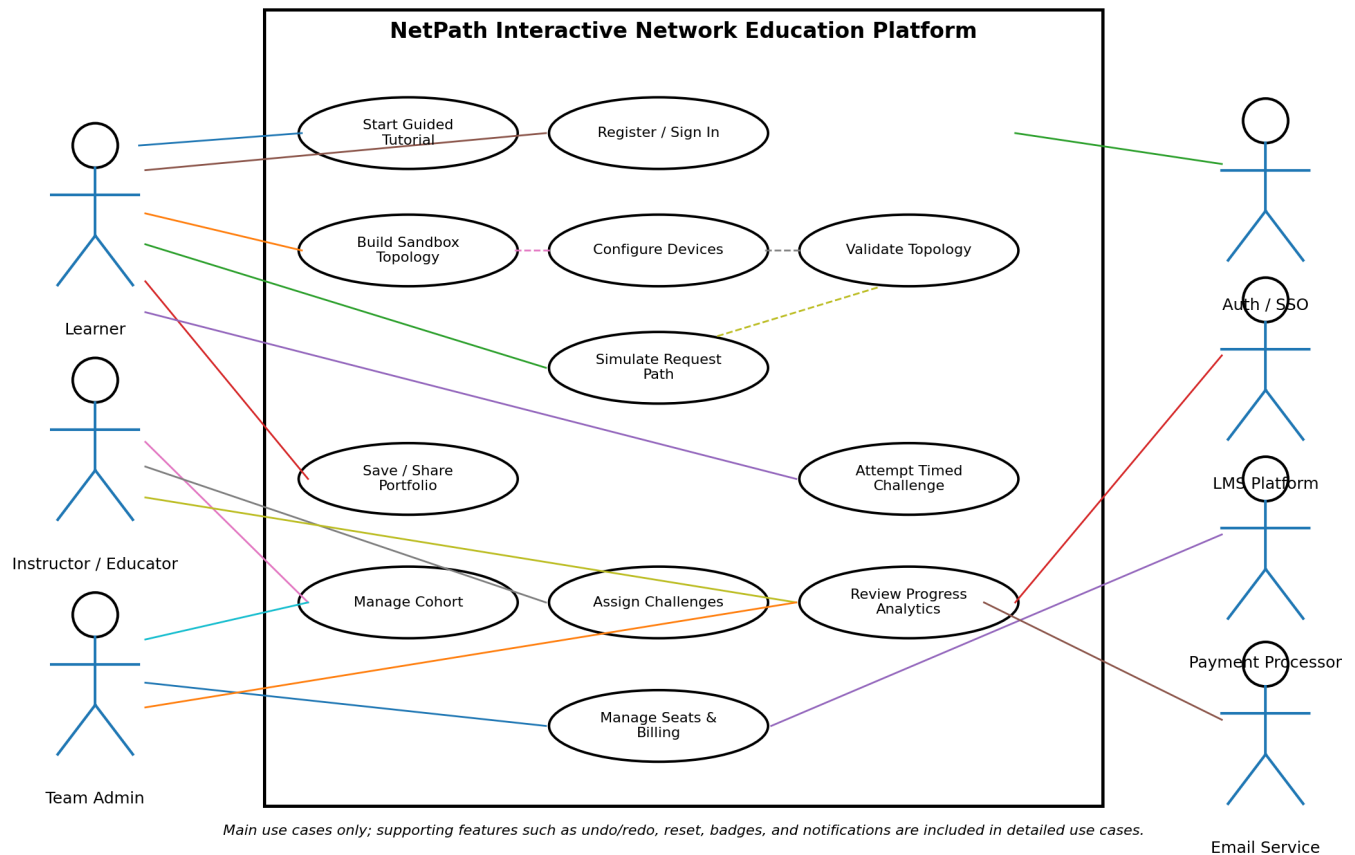
Actor	ID	Use Case	High-Level Description
Curriculum / Platform Administrator	UC-A01	Author Curriculum Stage	Create or update guided lessons, diagrams, hints, and practice tasks.
Curriculum / Platform Administrator	UC-A02	Define Validation Rules	Configure the success criteria used by the validation engine for each stage or challenge.
Curriculum / Platform Administrator	UC-A03	Publish Content Module	Release modules, unlock device cards, and manage content availability.
Curriculum / Platform Administrator	UC-A04	Monitor Product Analytics	Analyze usage, completion, retention, and conversion metrics.

4. Main Use Case Summaries

ID	Use Case	Actors	Preconditions	Basic Flow	Outcome
UC-L01	Start Guided Tutorial	Learner	Learner selects an available stage.	System displays explanation, annotated diagram, and practice activity.	Learner understands the concept and reaches the interactive task.
UC-L03	Build Sandbox Topology	Learner	Learner is in a guided stage or challenge.	Learner drags devices onto canvas and draws links between ports or wireless endpoints.	A topology graph is created or updated.
UC-L04	Configure Devices	Learner	A topology contains configurable devices.	Learner selects a device and enters stage-appropriate values such as IP address, gateway, or DHCP setting.	Device configuration is stored and made available to validation.
UC-L05	Validate Topology	Learner; Validation Engine	Topology or device configuration changes.	System checks reachability and lesson success criteria and displays color-coded feedback.	Learner knows whether the network is correct, partially correct, or broken.
UC-L06	Simulate Request Path	Learner; Simulation Engine	Topology has passed validation.	Learner clicks Send Request; system animates the outbound request and return path with explanatory tooltips.	Learner sees how traffic moves through the network conceptually.
UC-E02	Assign Lessons and Challenges	Instructor / Educator	Instructor has a cohort.	Instructor selects stages or challenges, sets due dates or assignment settings, and publishes to learners.	Learners receive assigned work.
UC-E03	Review Progress Analytics	Instructor / Educator; Team Admin	Learners have completed activity.	Actor opens dashboard and reviews completions, stage results, and engagement metrics.	Actor identifies learner progress and training outcomes.
UC-T02	Manage Seats and Billing	Team Administrator; Payment Processor	Team or institution wants paid access.	Administrator selects plan, manages seats, and completes billing workflow.	Team subscription and access are updated.

5. Use Case Diagram - Main Use Cases

The diagram below shows the main use cases and the primary relationships between the actors and the NetPath system boundary.



6. Key Assumptions and Notes

- Learners can begin early tutorial content without signing up, but registration is required to save progress after initial activation milestones.
- The validation engine is treated as an internal supporting system rather than an external actor, because it is part of the NetPath product behavior.
- Authentication, payment, email, and LMS services are secondary actors because they support core user goals but do not initiate learning workflows.
- The initial use case scope covers beginner-first LAN and WAN curriculum, sandbox topology creation, simulation, progress tracking, and team or academic administration.
- Advanced future capabilities such as AI-powered hints, community challenge marketplace, and white-label API embedding can be modeled as later-phase use cases.